YUNSUNG CHUNG

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RESEARCH PROFILE

Ph.D. candidate in Computer Science at Tulane University, expected to graduate in December 2026. Research focuses on longitudinal EHR modeling, uncertainty and evaluation for clinical AI, wearable biosignals, and clinically aligned generative models. Current first-author work includes submissions to KDD 2027, NeurIPS 2026, and MICCAI 2026, alongside peer-reviewed publications at USENIX Security, MICCAI, ICASSP, and AI-ML Systems.

EDUCATION

Tulane University *Aug 2022 – Present*
Ph.D. in Computer Science, expected Dec 2026 — New Orleans, LA

The Australian National University *Jul 2019 – Jul 2021*
M.S. in Computer Science — Canberra, Australia

Dongguk University *Mar 2012 – Feb 2018*
B.S. in Electronics, Information & Communication Engineering — Seoul, South Korea

EXPERIENCE

Samsung Research America – Research Intern *May 2025 – Aug 2025*
Mountain View, CA — Supervisor: Dr. Sharanya Desai

- Designed a biomarker-specific temporal decay loss for smartwatch PPG paired with sparse clinical labels. Across 450 participants and 10 biomarkers, the model averaged 0.712 AUROC and 0.715 AUPRC.
- Built the Python/PyTorch pipeline for signal filtering, quality screening, temporal windowing, training, subject-wise validation, ablations, and reproducible reporting.
- Worked with research scientists and engineers to turn wearable sensing constraints into a fixed modeling and evaluation protocol.

Tulane University, TRIAD Lab – Research Assistant *Aug 2022 – Present*
New Orleans, LA — Supervisor: Dr. Jihun Hamm

- Lead research on longitudinal EHR modeling, wearable biosignals, medical imaging, and synthetic data with collaborators in computer science, cardiology, and industry.
- Developed audits that separate fixed-anchor prediction from long-horizon trajectory modeling, including future-state recovery, uncertainty, robustness, and failure-mode tests for clinical foundation models.
- Built EHRDyn, a five-task benchmark and public artifact for comparing 21 offline-RL method specifications and six OPE estimators across complementary simulators, with analyses of estimator support, uncertainty, and ranking stability.
- Developed CRAFT, a reward-based diffusion adaptation method that uses clinical checklists and multimodal model feedback to improve medical image alignment and downstream utility.
- Developed multimodal models for atrial-fibrillation ablation outcomes using LGE-MRI, procedure variables, and follow-up data; published the SOFA framework at MICCAI 2025.

Airport Railroad Express – Hardware Engineer *Nov 2017 – May 2019*
Incheon, South Korea

- Tested and debugged automated fare-collection and passenger-service hardware in a high-reliability transportation system.

SELECTED RESEARCH CONTRIBUTIONS

- **EHR foundation-model audit:** Tests whether fixed-anchor outcome gains transfer to long-horizon trajectory structure, future-state recovery, and uncertainty.
- **EHRDyn benchmark:** Compares forecasting, offline-RL, world-model, and OPE workflows under fixed contracts and complementary simulators.
- **Clinically aligned generation:** Uses structured clinical criteria and multimodal evaluators to train and audit medical diffusion models.
- **Sparse-label wearable learning:** Learns biomarker-specific temporal decay from dense PPG and infrequent laboratory labels.

SELECTED PUBLICATIONS

[1] **Yunsung Chung**, Han Feng, Nassir Marrouche, and Jihun Hamm. *EHRDyn: A Benchmark for Offline Reinforcement Learning from EHRs with Complementary Simulators*. Under review at **KDD**, 2027.

[2] **Yunsung Chung**, Han Feng, Nassir Marrouche, and Jihun Hamm. *Auditing Long-Horizon Patient Trajectory Modeling in EHR Foundation Model Benchmarks*. Under review at **NeurIPS**, 2026.

[3] **Yunsung Chung**, Alex El Darzi, Carlo El Khoury, Han Feng, Nassir Marrouche, and Jihun Hamm. *CRAFT: Clinical Reward-Aligned Finetuning for Medical Image Synthesis*. Under review at **NeurIPS**, 2026.

[4] **Yunsung Chung**, Yingshuo Liu, Abboud F. Hassan, Han Feng, Mary M. Maleckar, Nassir Marrouche, and Jihun Hamm. *Intervention-Aware Clinical World Model for Post-OP Outcome Forecasting in Cardiology*. Under review at **MICCAI**, 2026.

[5] **Yunsung Chung**, Keum San Chun, Migyeong Gwak, Han Feng, Yingshuo Liu, Chanho Lim, Viswam Nathan, Nassir Marrouche, and Sharanya Arcot Desai. *Weighted Temporal Decay Loss for Learning Wearable PPG Data with Sparse Clinical Labels*. **ICASSP**, 2026.

[6] **Yunsung Chung**, Chanho Lim, Ghassan Bidaoui, Christian Massad, Nassir Marrouche, and Jihun Hamm. *SOFA: Deep Learning Framework for Simulating and Optimizing Atrial Fibrillation Ablation*. **MICCAI**, 2025.

[7] **Yunsung Chung**, Yunbei Zhang, Nassir Marrouche, and Jihun Hamm. *SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy of Synthetic Image Generation*. **USENIX Security**, 2025.

SERVICE AND HONORS

Reviewer: NeurIPS 2026; ICML 2026; MICCAI 2024–2026; IEEE TCSVT, 2023.
Honors: FIRA RoboWorld Cup 2016, 1st Place; International Robot Contest 2016, 2nd Place.

SKILLS

Methods: Longitudinal modeling, time-series forecasting, representation learning, multimodal learning, diffusion models, uncertainty estimation, model auditing.
Engineering: Python, PyTorch, NumPy, pandas, Scikit-learn, Hugging Face, SQL, Git, AWS, Weights & Biases.
Languages: Korean (native), English (fluent).